Hüseyin Felek

R&D Engineer | Industrial Automation & Process Instrumentation

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SUMMARY

Engineer with nearly 5 years in industrial automation and process instrumentation, carrying products and projects from requirement definition through design, testing, documentation, handover to production and first field trials. Owned a new product end to end, rolled out a redesigned controller architecture to 1000+ field devices, and coordinated with customers, production, quality and field service teams, interns and an international technology vendor. Hands-on with PLC and industrial communication (CODESYS, PROFINET, Modbus), production testing and quality assurance, and project tracking in Jira and Confluence.

WORK HISTORY

Software Design Engineer · Andasis Elektronik 05.2025 – 02.2026

- Delivered core network features (VLAN, IEEE 802.1X/RADIUS, ACL) for an OpenWrt-based switching and routing platform to production readiness, troubleshooting reliability issues at packet level with Wireshark.

Embedded Systems R&D Engineer · Eliar Elektronik 07.2021 – 05.2025

Project and Product Delivery

- Defined requirements and specifications for a new dual-channel pH/ORP product based on customer needs and owned it end to end, coordinating design, firmware, documentation and certification activities through to production readiness.
- Redesigned the I/O architecture of the company's automation controllers and rolled it out to 1000+ field devices, replacing legacy controllers.
- Coordinated integration with an international PROFINET stack vendor, managing technical alignment and issue resolution.
- Took part in the import process for pH probes sourced from abroad, supporting the delivery of components needed for production.
- Defined compliance tests and ran certification testing for the company's first officially BLE-certified device.
- Tracked tasks and maintained project documentation in Jira and Confluence; coordinated interns.

Handover, Testing and Quality

- Defined production test procedures, manufacturing guidelines and post-production quality criteria, and handed production tests and calibration procedures over to the production team.
- Built CODESYS-based and PC-based test and calibration stations with automated test sequencing and data logging, shortening test and calibration time in series production.
- Took part in first field trials of new devices, carried out post-delivery checks and resolved defects found after delivery; led post-production quality assurance for the controller platform.

Customer and Field Interface

- Led a 6-month on-site feasibility study at customer dyehouses: interviewed process teams, collected samples, analyzed data with Python, SQL and Power BI, and presented the findings to management.
- Investigated issues encountered on field devices down to their technical causes, trained the field service team on calibration procedures and served as the technical escalation point for customer and field issues, responding to customers directly.

Automation and Industrial Communication

- Architected an embedded Linux automation controller platform with software-defined variants, from a PROFINET I/O coupler to a full CODESYS PLC.
- Developed IEC 61131-3 applications in CODESYS and device drivers for I/O module data exchange over Modbus and PROFINET.

Intern · Siemens (Mobility) 07.2018 – 09.2018

CORE COMPETENCIES

Project Coordination	Requirement and scope definition, task and schedule tracking (Jira), vendor coordination, customer interface, project documentation (Confluence), Scrum
Testing & Quality	Production test procedures, test and calibration stations, quality criteria, certification testing, root cause analysis
Industrial Automation	PLC programming (CODESYS, IEC 61131-3), PROFINET, Modbus RTU/TCP, IO-Link, 4-20 mA, process instrumentation (pH/ORP, conductivity, optical)
Office & Data Tools	MS Office (Excel, PowerPoint), Power BI, SQL, Python
Software	C/C++, embedded Linux, Docker, Git

EDUCATION & LANGUAGES

B.Sc., Electrical and Electronics Engineering · Boğaziçi University 2016 – 2021

Turkish Native | English Advanced | French Intermediate

Driving licence Class B | Willing to travel internationally